The Affordability Squeeze

How Rising Costs Outpaced Income (2009–2023)

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Overview

The Question

Between 2009 and 2023, the U.S. CPI rose ~42%, reshaping household budgets nationwide.

Did income growth keep pace — and did the burden fall equally across all communities?

Data Sources

- U.S. Census ACS (2009–2023)
  - 3,000+ county-level observations per year
  - Income, rent, home costs, poverty rates
- BLS CPI-U (2000–2024)
  - Monthly CPI aggregated to yearly averages
  - Used to adjust all dollar figures to 2024 $

Key Finding: High-poverty counties experienced meaningfully lower inflation-adjusted income growth, widening the gap with wealthier communities over the study period.

Source: U.S. Census Bureau ACS | Bureau of Labor Statistics CPI-U

Data Pipeline & Key Transformations

01

Load & Clean

Standardized FIPS codes with str_pad(). Removed duplicates and impossible values. Filtered to 2009–2023.

02

CPI Aggregation

Parsed date strings with lubridate::ymd(). Averaged 12 monthly CPI-U values into annual averages.

03

Joins

Merged census county-year data to CPI yearly averages. All 45,000+ rows preserved.

04

Inflation Adjust

Created 2024-dollar versions of income, rent, and home cost using the CPI ratio formula.

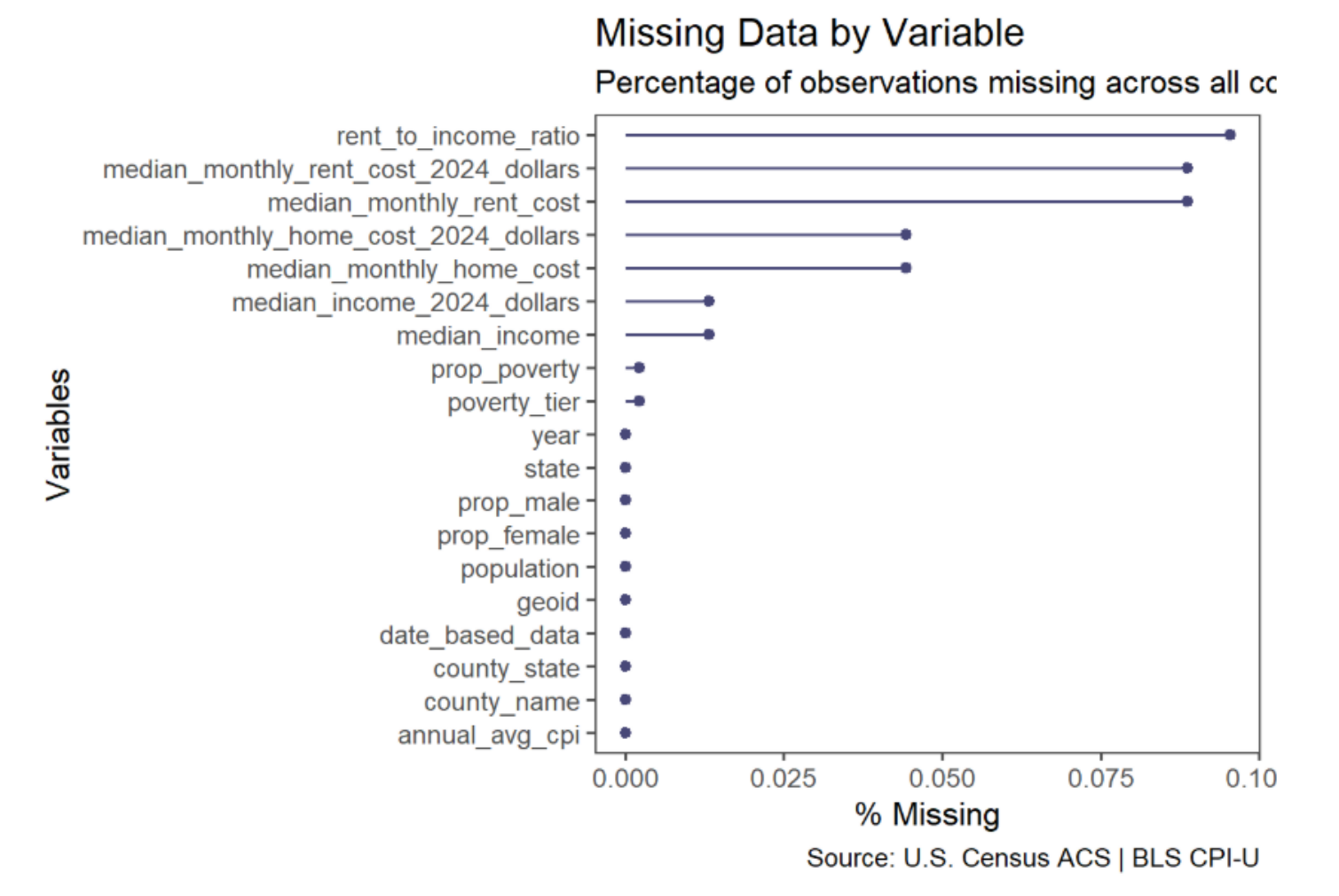
05

Derived Variables

Extracted state and county via str_extract(). Assigned poverty quartile tiers with ntile(). Computed rent-to-income ratio.

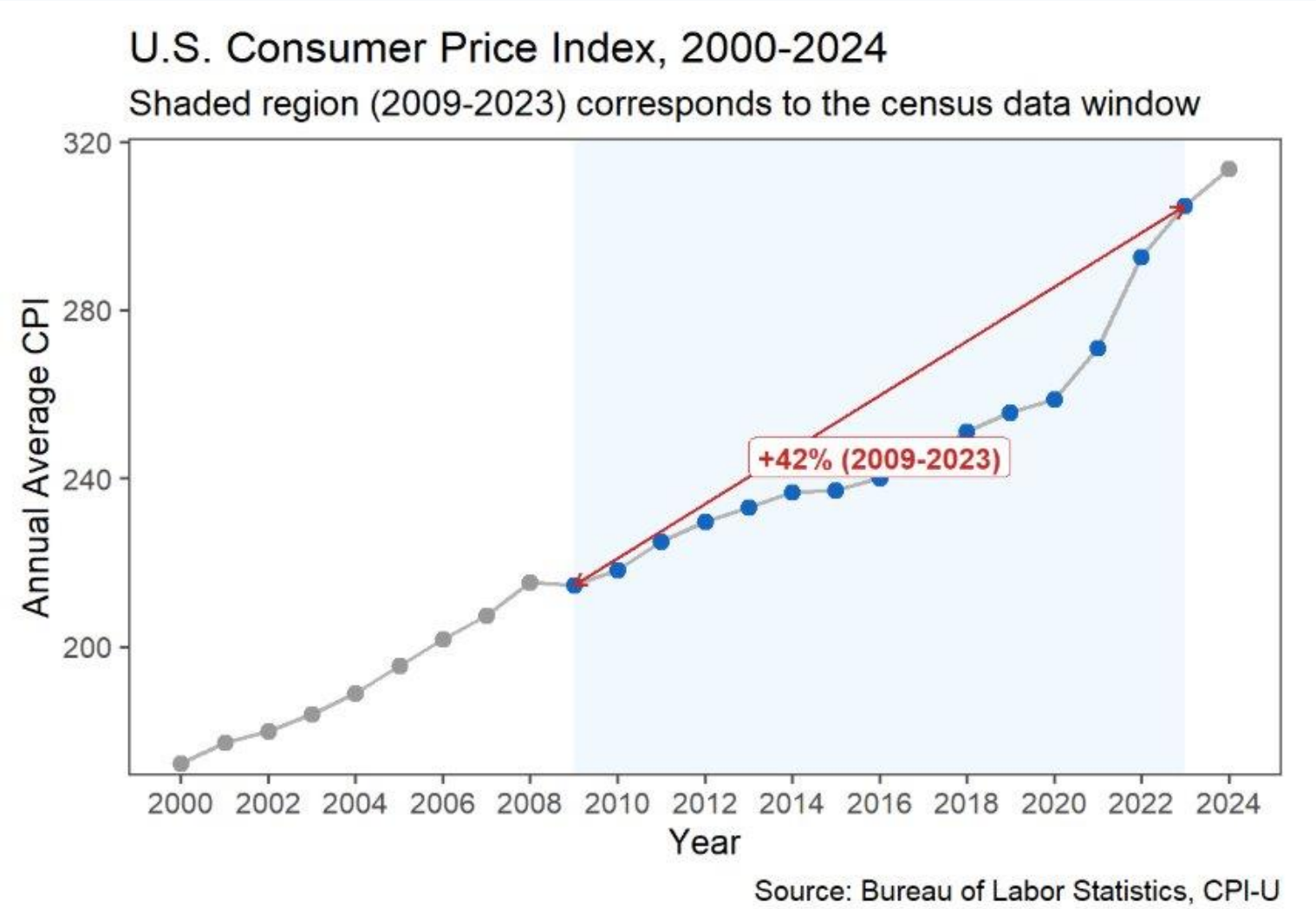
stringr: str_pad, str_squish, str_detect, str_extract, str_remove | lubridate: ymd(), year()

Patterns of Missingness



Visualized with naniar::gg_miss_var().

Section 1: The Inflation Context



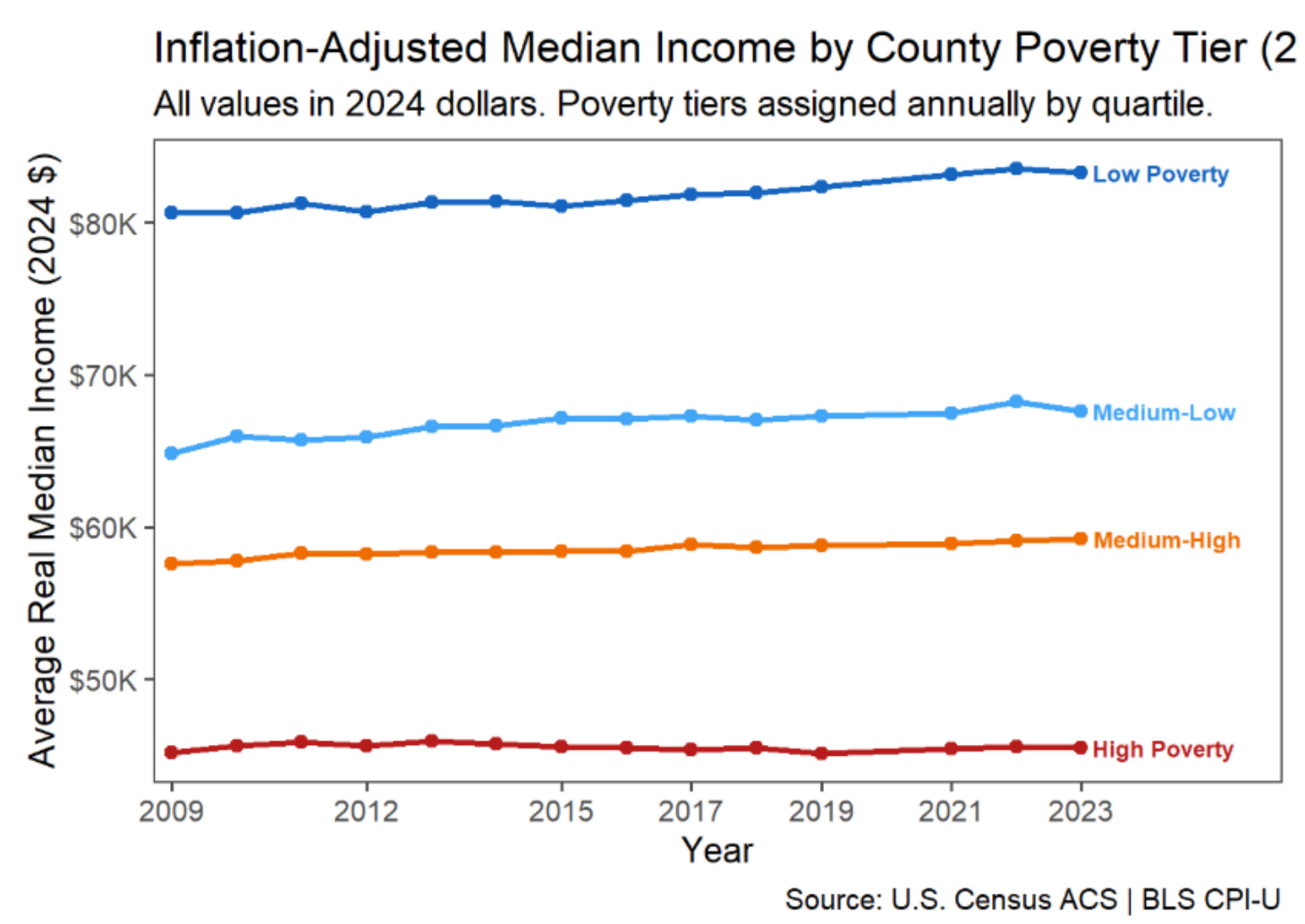
+42%
CPI increase
2009–2023

214 → 305
CPI index
points

Post-2020
Inflation
accelerated

U.S. CPI trend from 2000 to 2024. The shaded region marks the 2009-2023 study window. Source: Bureau of Labor Statistics.

Section 2: Did Incomes Keep Up?



Key Takeaway

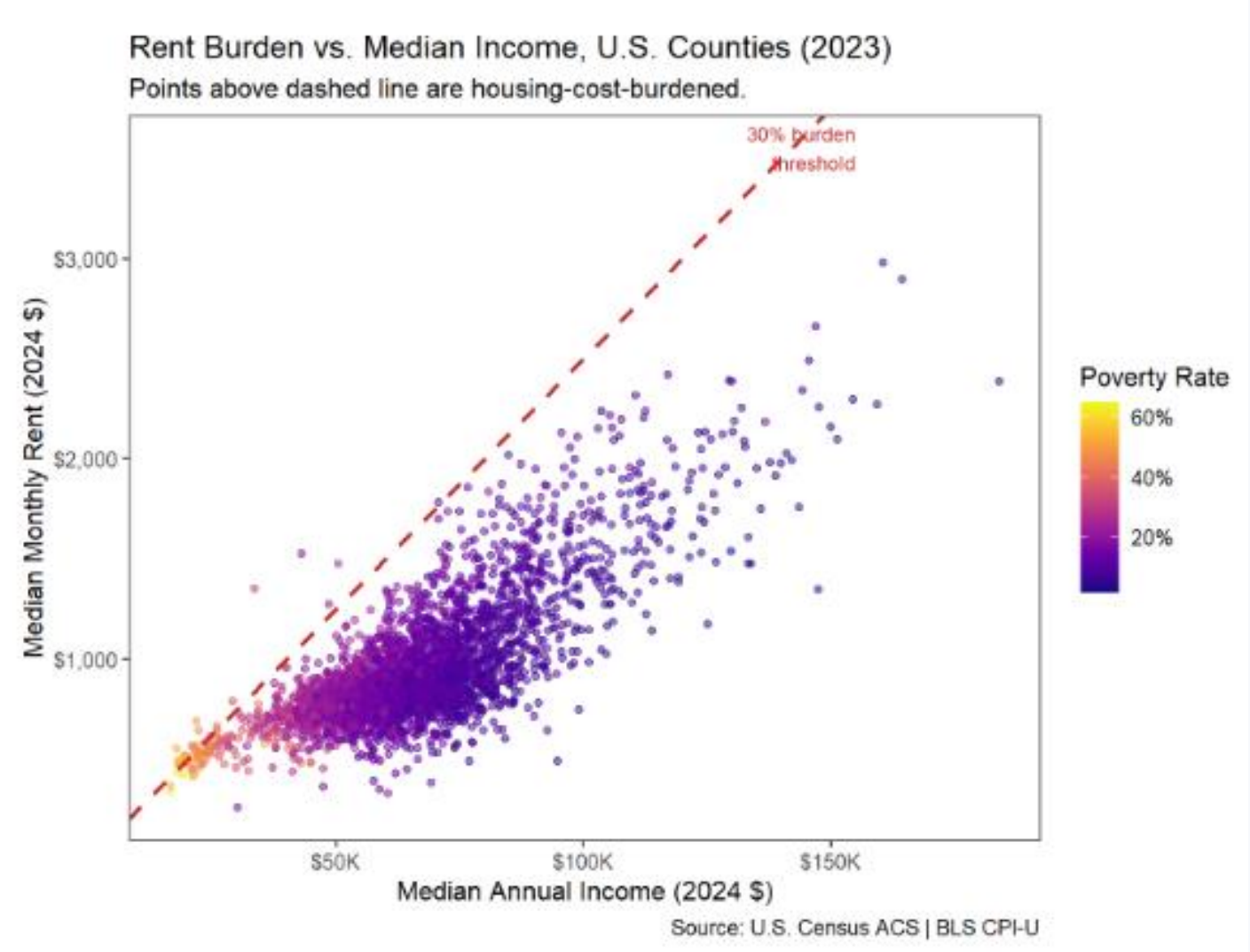
- All tiers grew in real terms — but the gap widened.
- Low Poverty counties surged post-2020 as labor markets tightened.
- High Poverty counties saw minimal gains and ended 2023 further behind in real terms than they started.

The squeeze was not shared equally.

Section 3: The Housing Burden

The 30% Rule

Households spending more than 30% of gross income on rent are considered housing-cost-burdened.

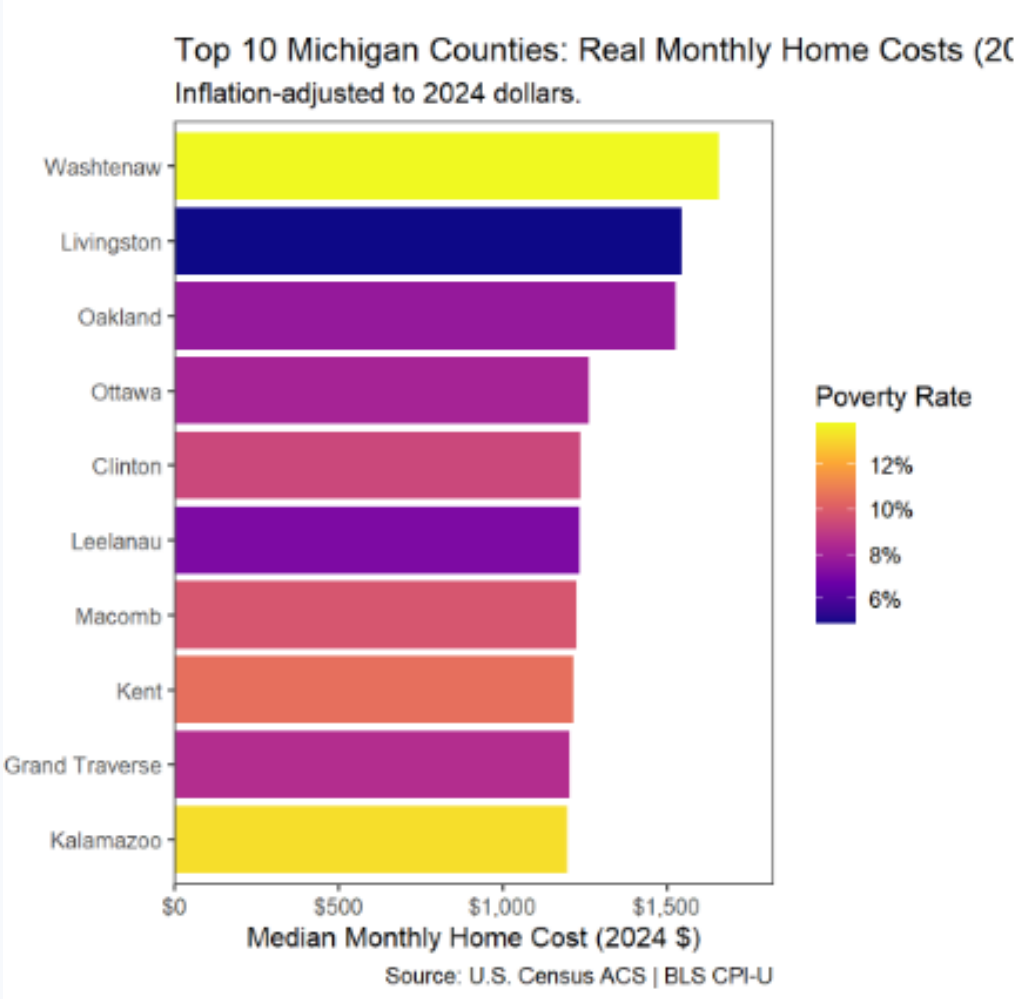


% of counties above the 30% rent-to-income threshold by poverty tier. Source: U.S. Census ACS | BLS CPI-U. 2023 data.

Section 3: The Housing Burden

Michigan

- Michigan's most expensive counties — Ottawa, Washtenaw, Leelanau — are among the state's lowest-poverty areas.
- High housing costs reflect rising market prices, not income deprivation.
- **This is a different housing crisis from what high-poverty counties face.**

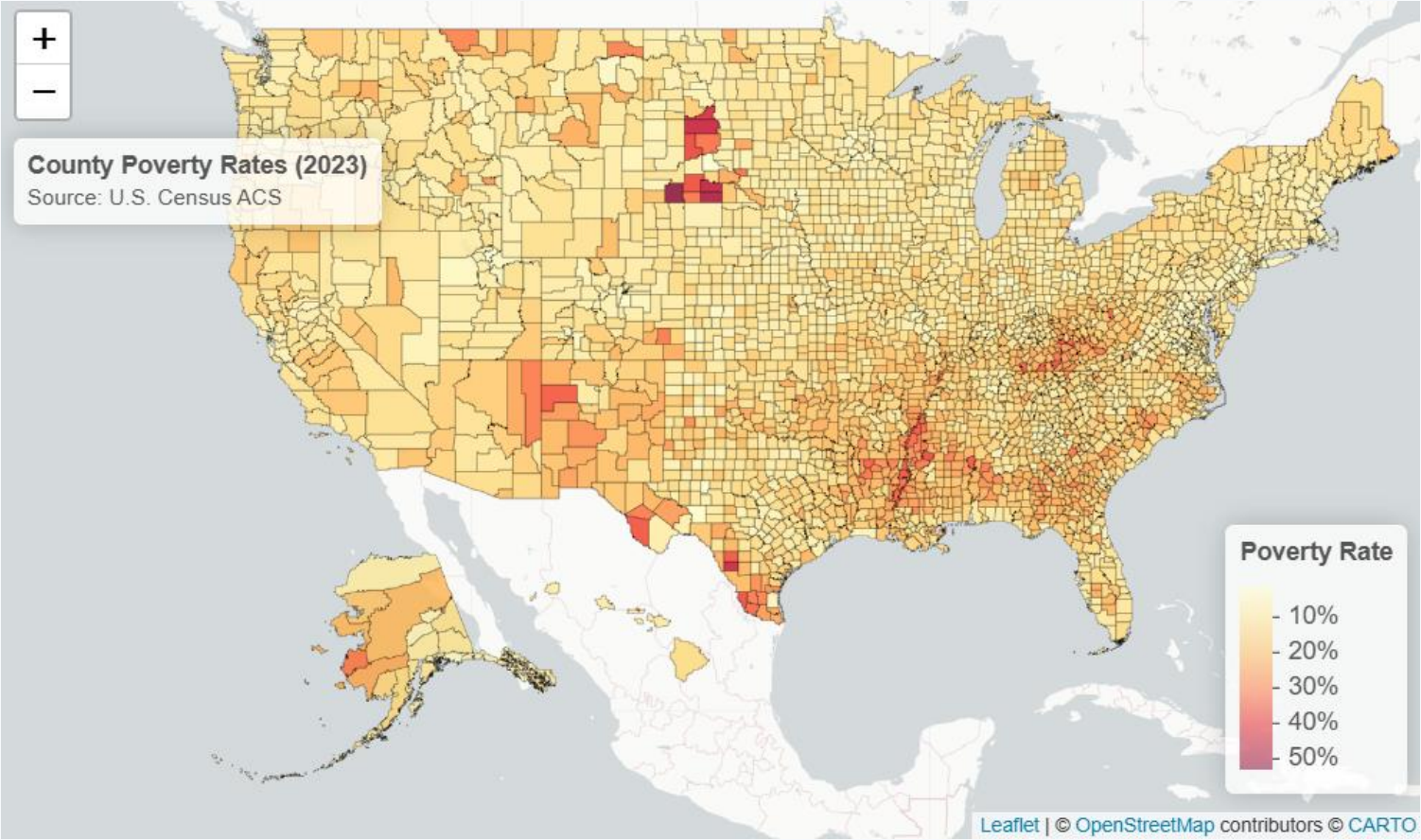


% of counties above the 30% rent-to-income threshold by poverty tier. Source: U.S. Census ACS | BLS CPI-U. 2023 data.

Section 4: Where It Hurts Most

Two Different Geographies of Hardship

- **MAP A: Poverty Rate (2023)**
- Concentrated in some states
- Persistent, structural poverty regions where incomes have chronically lagged behind national averages.

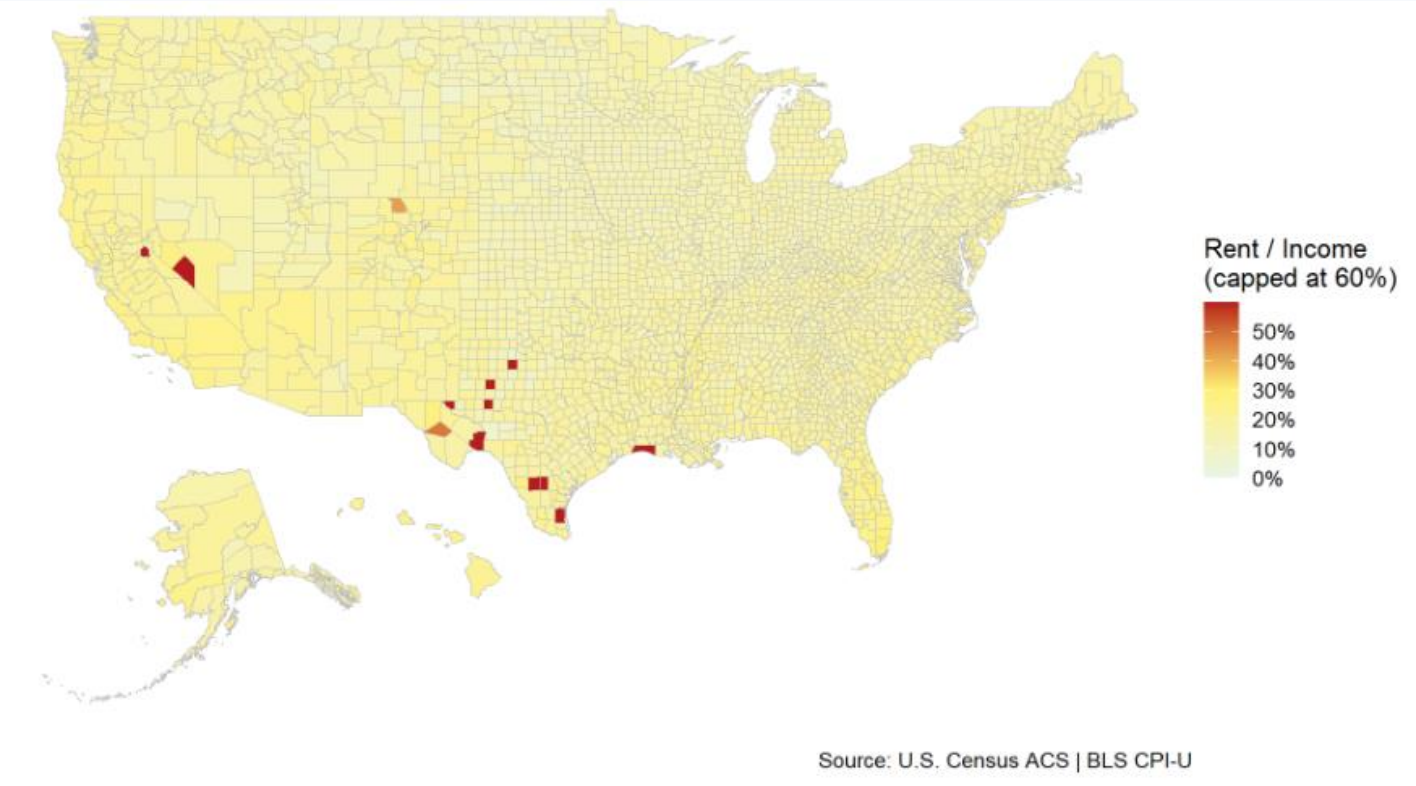


Interactive Leaflet maps in the HTML report. Static choropleth for rent burden. Source: U.S. Census ACS | BLS CPI-U.

Section 4: Where It Hurts Most

Two Different Geographies of Hardship

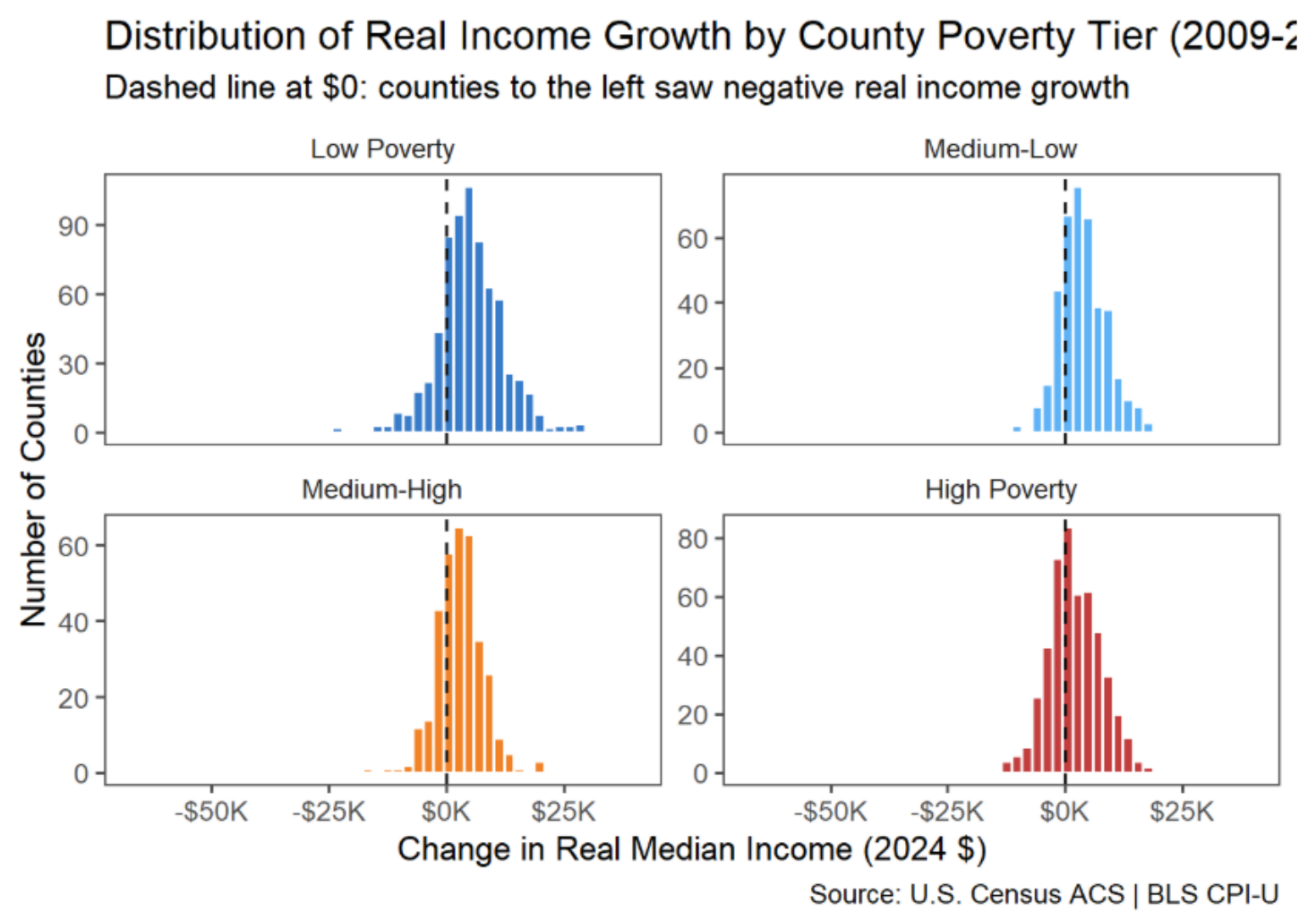
- **MAP B: Rent-to-Income Ratio (2023)**
- Concentrated in:
  - Coastal metros
  - Mountain resort counties
  - High-demand urban areas
- A different crisis: rent outpaces income here regardless of poverty rate; driven by supply constraints and market demand.



Interactive Leaflet maps in the HTML report. Static choropleth for rent burden. Source: U.S. Census ACS | BLS CPI-U.

Section 5: Income Growth Distribution by Poverty Tier

Distributional view of real income growth (2009–2023) per county, split by poverty tier.



Permutation Test: Is the Gap Statistically Significant?

Hypotheses

H_0: Mean real income growth is the same for High and Low Poverty counties.

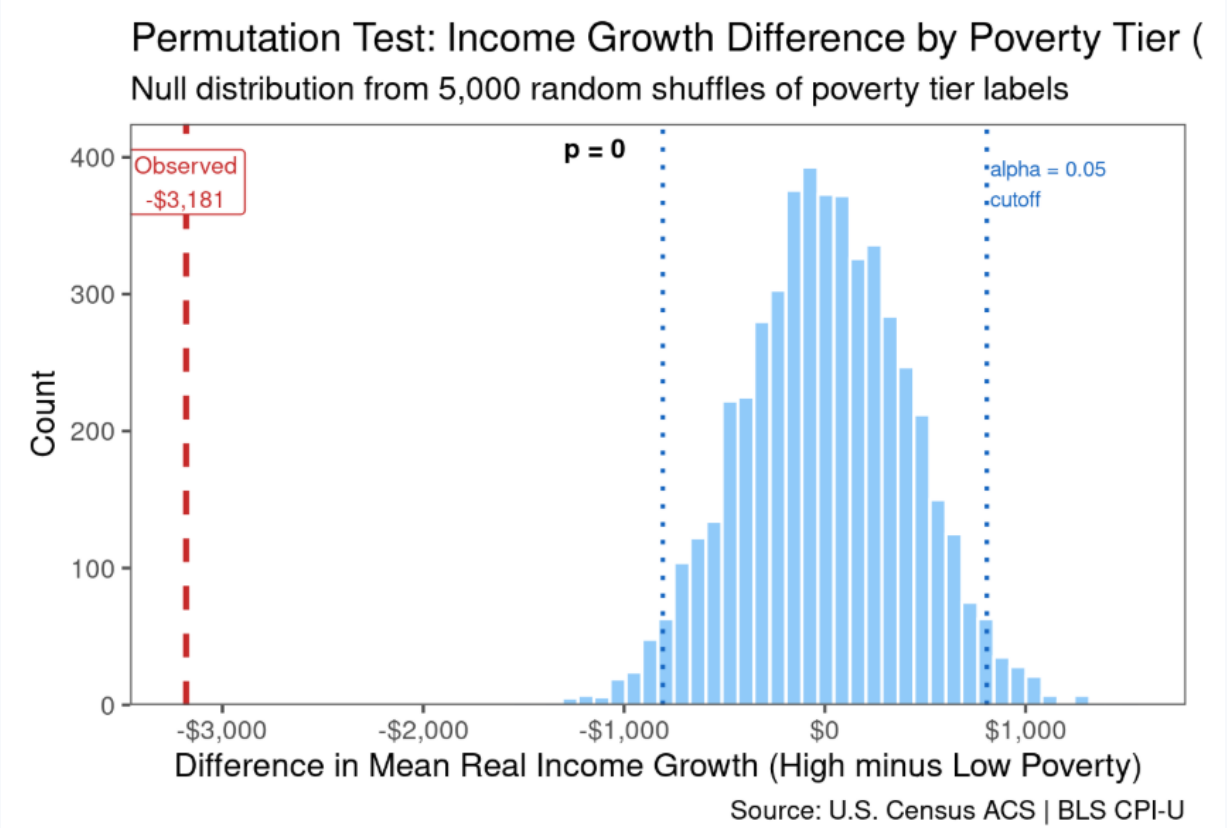
H_1: Mean real income growth differs between the two groups.

Results

5,000 permutations | Seed: 518

p-value < 0.0001

Reject H_0 — the gap is not due to chance.



← Observed gap far outside null distribution

Two-sided permutation test. Test statistic: mean(High Poverty growth) − mean(Low Poverty growth). Source: U.S. Census ACS | BLS CPI-U.

Conclusions & Takeaways

01 Prices rose 42% from 2009 to 2023, accelerating sharply after 2020.

02 Incomes grew in all tiers — but the gap between Low and High Poverty counties widened in real terms.

03 Housing cost burden is concentrated in high-poverty counties, with 67% above the 30% threshold.

04 Geography matters: poverty concentrates in the South and Appalachia; rent burden is a coastal phenomenon. Two distinct crises.

05 The income growth gap is statistically significant ($p < 0.0001$). The squeeze was systematically worse for the most vulnerable.

Appendix

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